ELASTIC LOAD BALANCER

An Elastic Load Balancer in Amazon Web Services is a service that automatically distributes incoming application traffic across multiple targets, such as EC2 instances, to ensure high availability, scalability, and fault tolerance.

Types of Elastic Load Balancer in Amazon Web Services

Application Load Balancer (ALB)

- Works at **Layer 7 (Application Layer)**
- Used for **web applications**
- **Protocols:** HTTP, HTTPS

Network Load Balancer (NLB)

- Works at **Layer 4 (Transport Layer)**
- Used for **high performance and low latency**
- **Protocols:** TCP, UDP, TLS

Classic Load Balancer (CLB)

- Works at **Layer 4 & Layer 7**
- Old/legacy type
- **Protocols:** HTTP, HTTPS, TCP, SSL

Gateway Load Balancer (GWLB)

- Works at **Layer 3 (Network Layer)**
- Used for **security appliances**
- **Protocols:** IP (uses GENEVE protocol)

OSI MODEL WITH PROTOCOLS

Application Layer

- Used for user interaction (apps)
- **Protocols:** HTTP, HTTPS, FTP, SMTP, DNS

Presentation Layer

- Data formatting, encryption, compression

- **Protocols/Formats:** SSL/TLS, JPEG, PNG, MP4

Session Layer

- Manages sessions/communication
- **Protocols:** NetBIOS, RPC

Transport Layer

- End-to-end communication, reliability
- **Protocols:** TCP, UDP

Network Layer

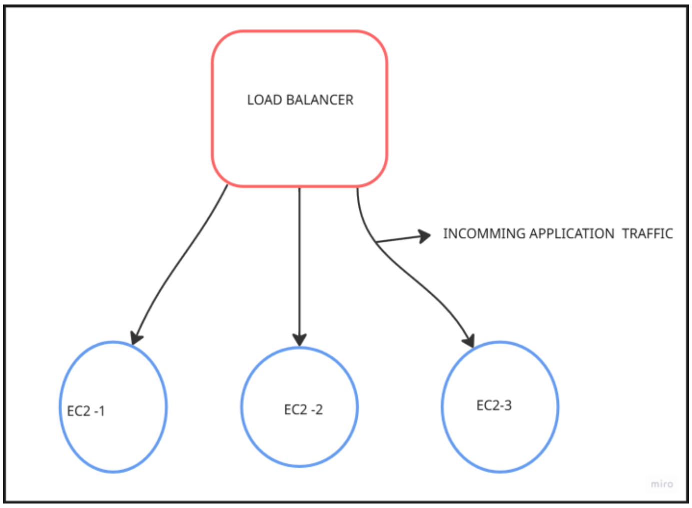
- Routing and IP addressing
- **Protocols:** IP, ICMP, IPSec

Data Link Layer

- Node-to-node transfer
- **Protocols:** Ethernet, ARP, PPP

Physical Layer

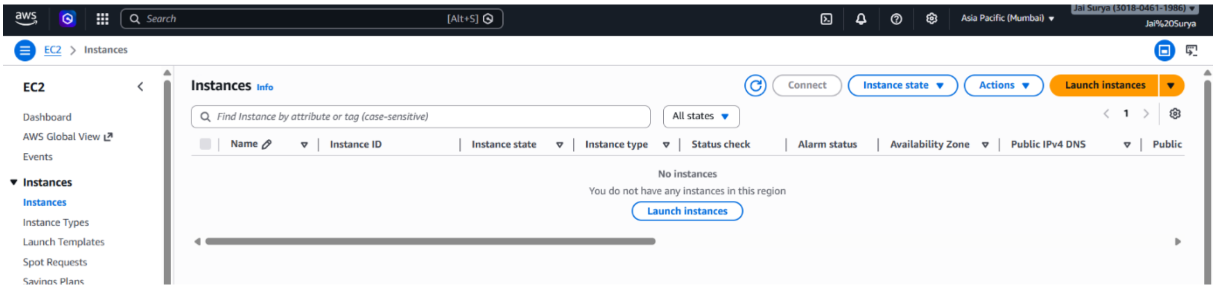
- Physical transmission of data
- **Examples:** Cables, Switch ports, Signals



HANDS-ON STEPS TO CREATE A CLASSIC LOAD BALANCER

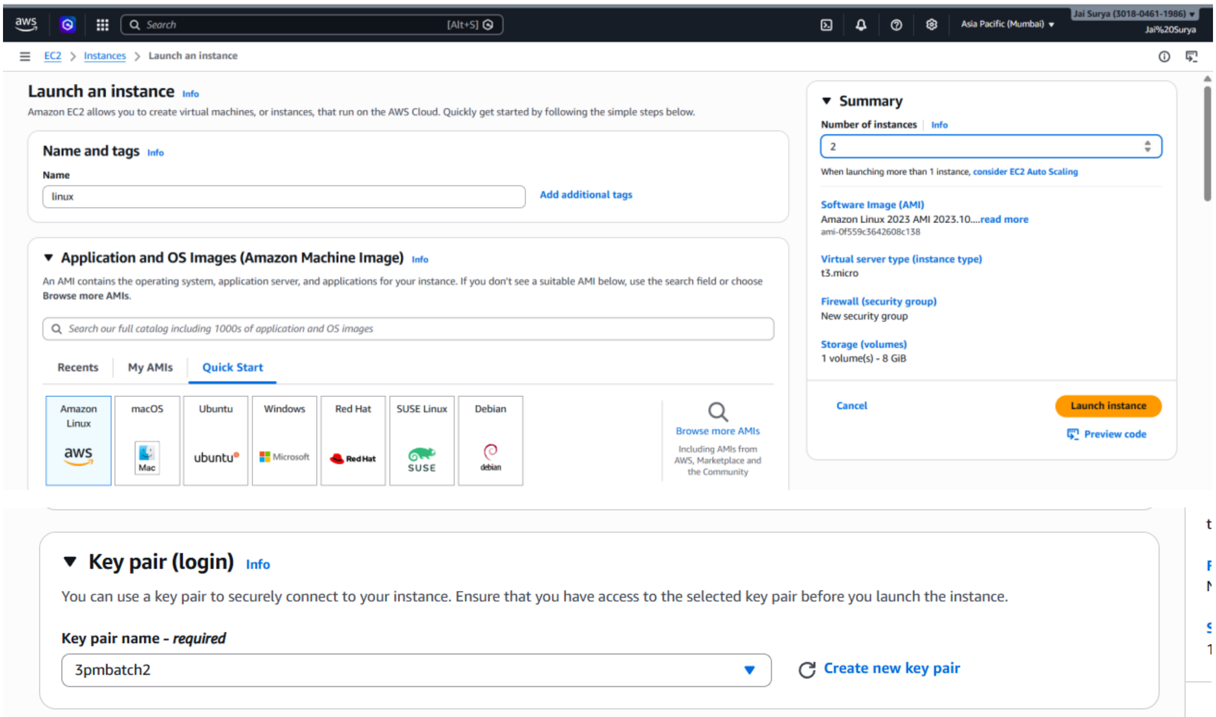
Screenshot: Launch Instance

- **Go to Amazon Web Services Console**
- **Open EC2 Dashboard**
- **Click Launch Instance**



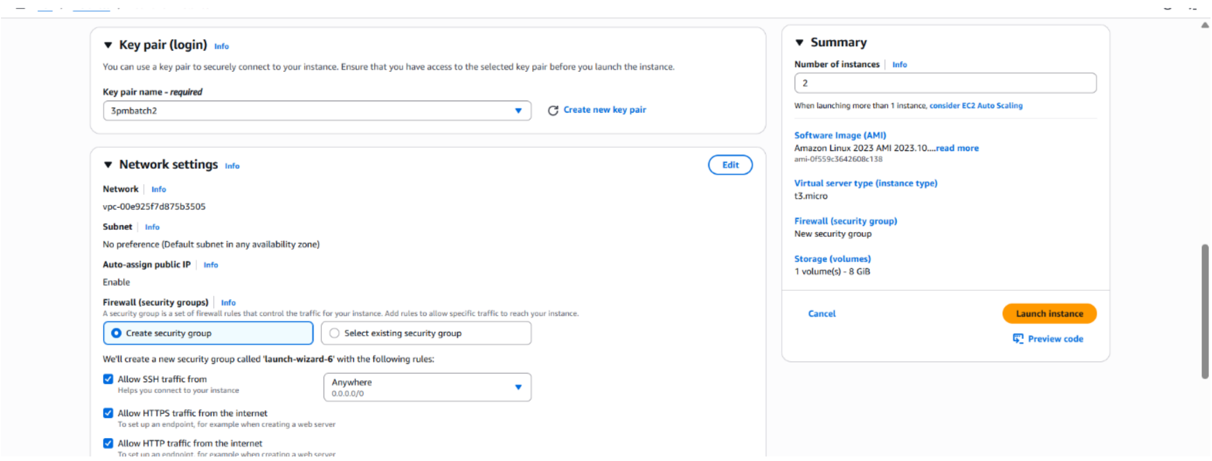
Screenshot: Create Instance

- Enter instance name
- Select Amazon Linux AMI
- Choose Key Pair
- Set number of instances to 2



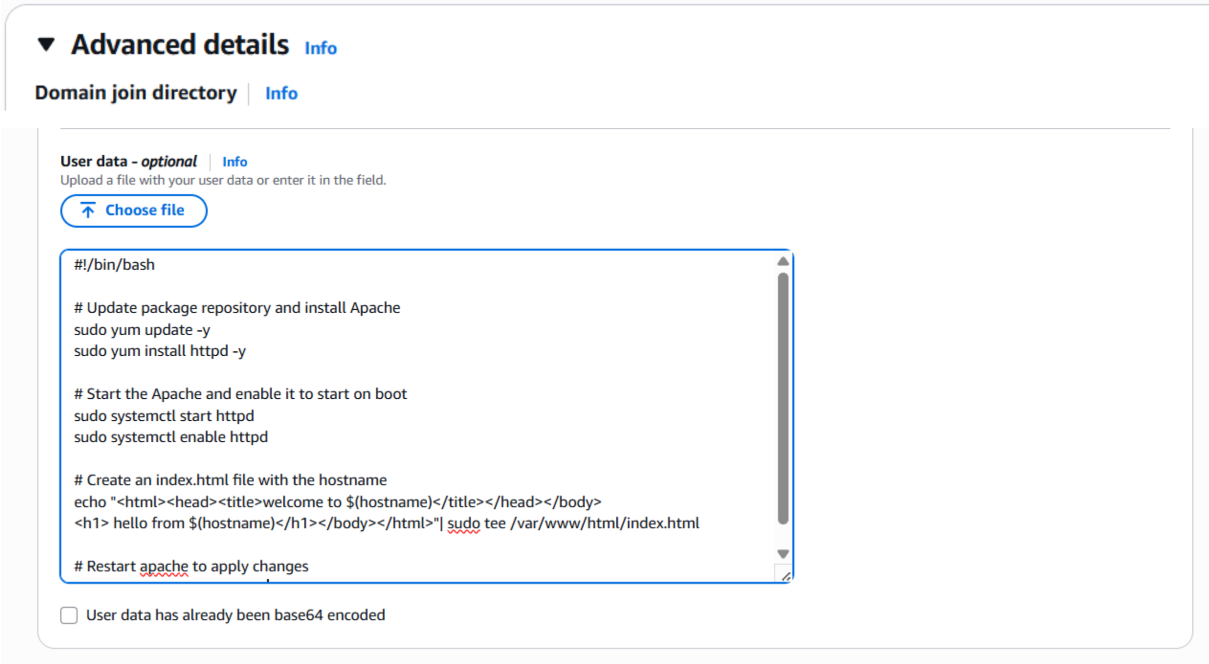
Screenshot: Configure Network Settings

- Allow SSH (port 22)
- Allow HTTP (port 80)
- Allow HTTPS (port 443)



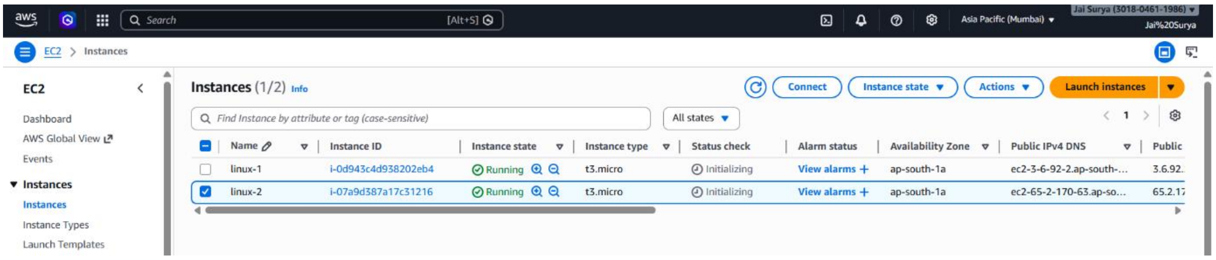
Screenshot: Advanced Details

- Go to Advanced Details
- Paste user data script in User Data
- Click Create / Launch Instance



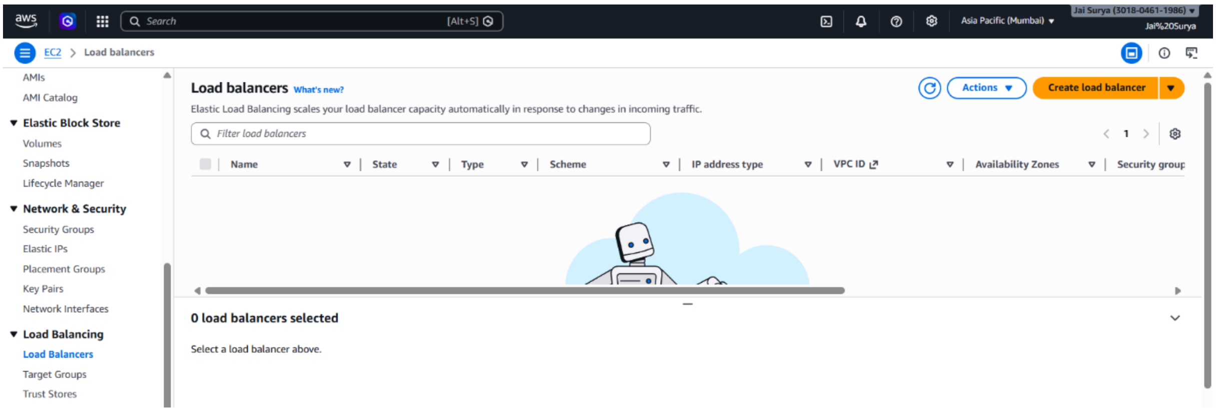
Screenshot: Instances Created

- Verify two instances are created
- Rename both instances



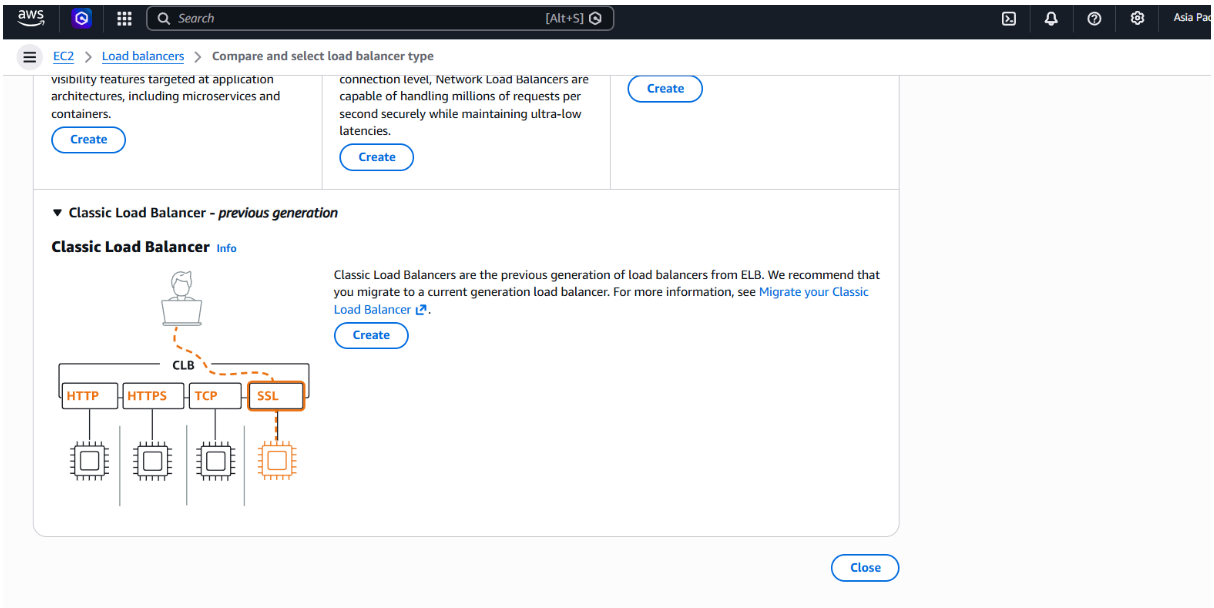
Screenshot: Go to Load Balancer

- Go to **Load Balancers**
- Click **Create Load Balancer**



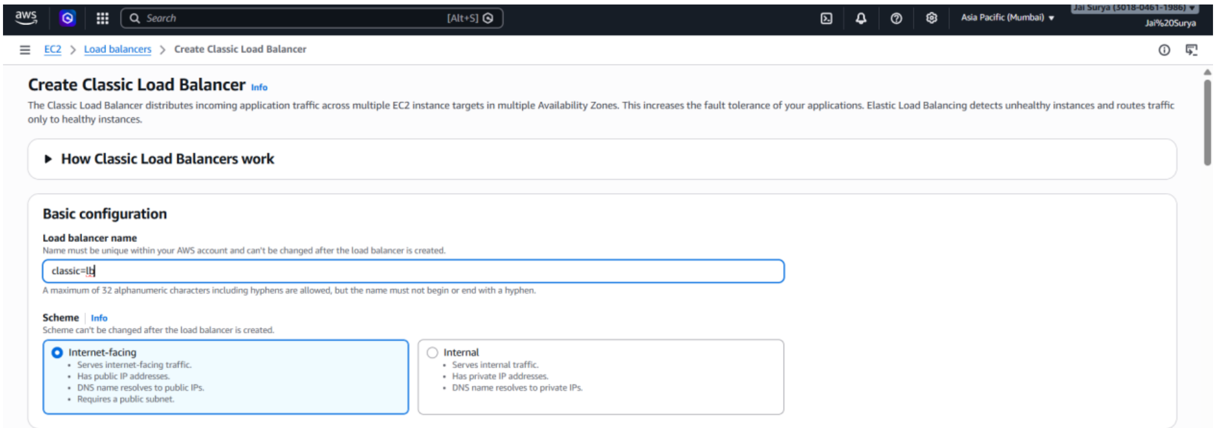
Screenshot: Select Classic Load Balancer

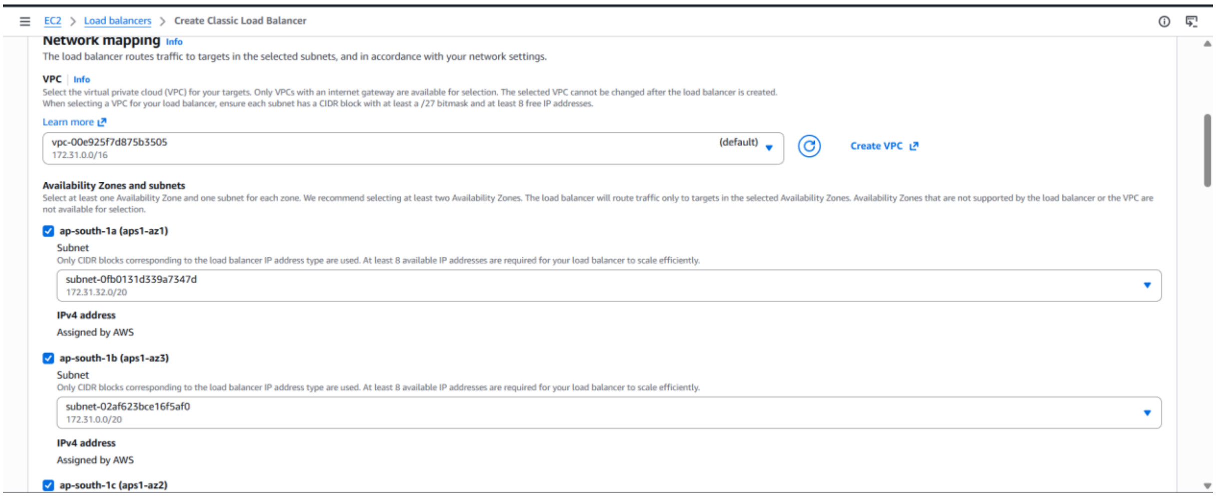
- Choose **Classic Load Balancer**
- Click **Create**



Screenshot: Configure Load Balancer

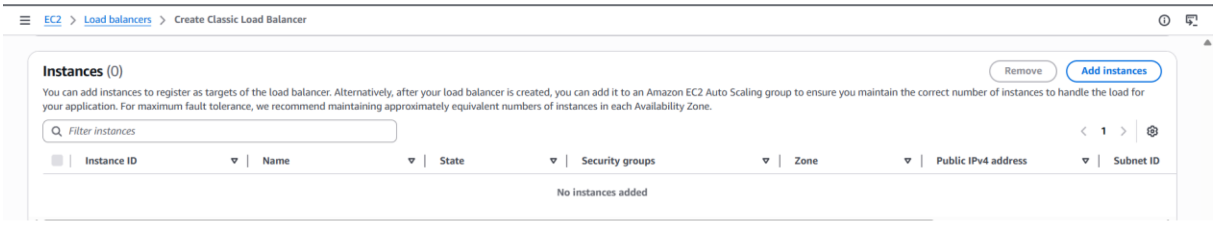
- Enter name
- Select Internet-facing
- Select all subnets





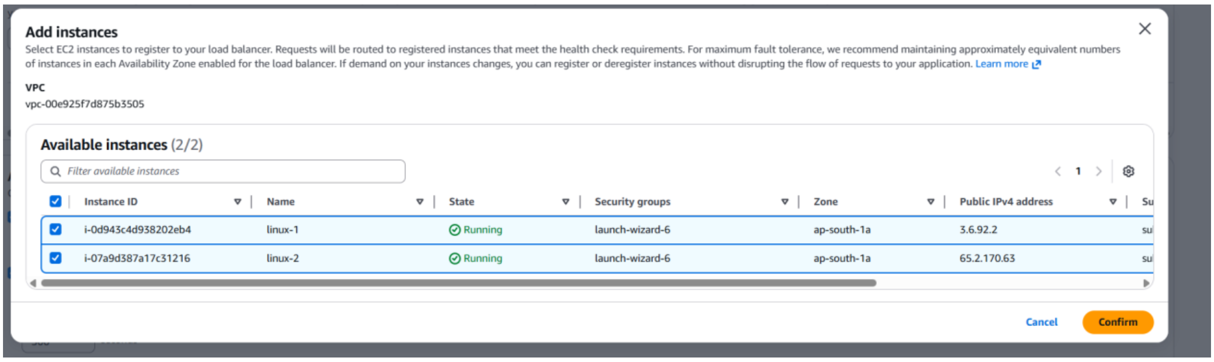
Screenshot: Add Instances

- Click **Add Instances**



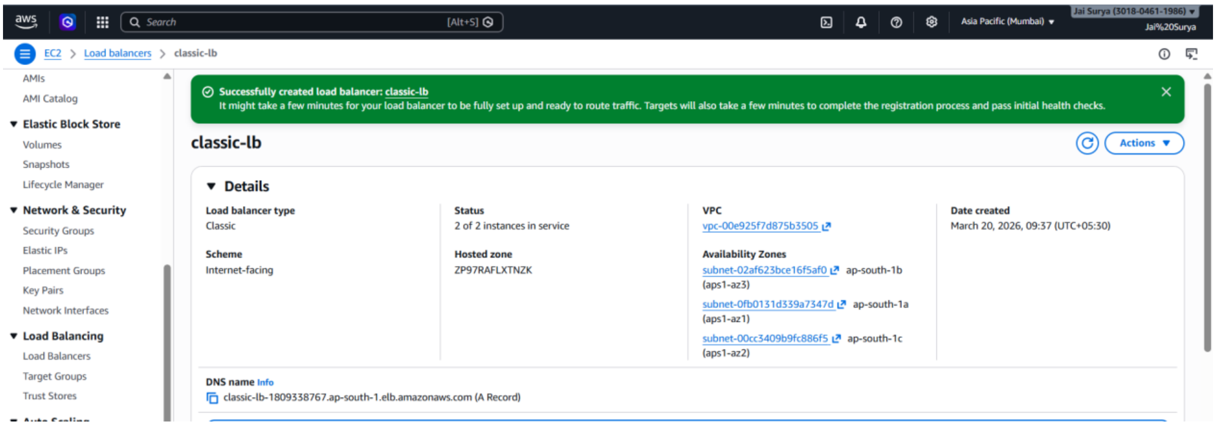
Screenshot: Select Instances

- Select the two created instances
- Click **Confirm / Add**



Screenshot: Copy DNS Name

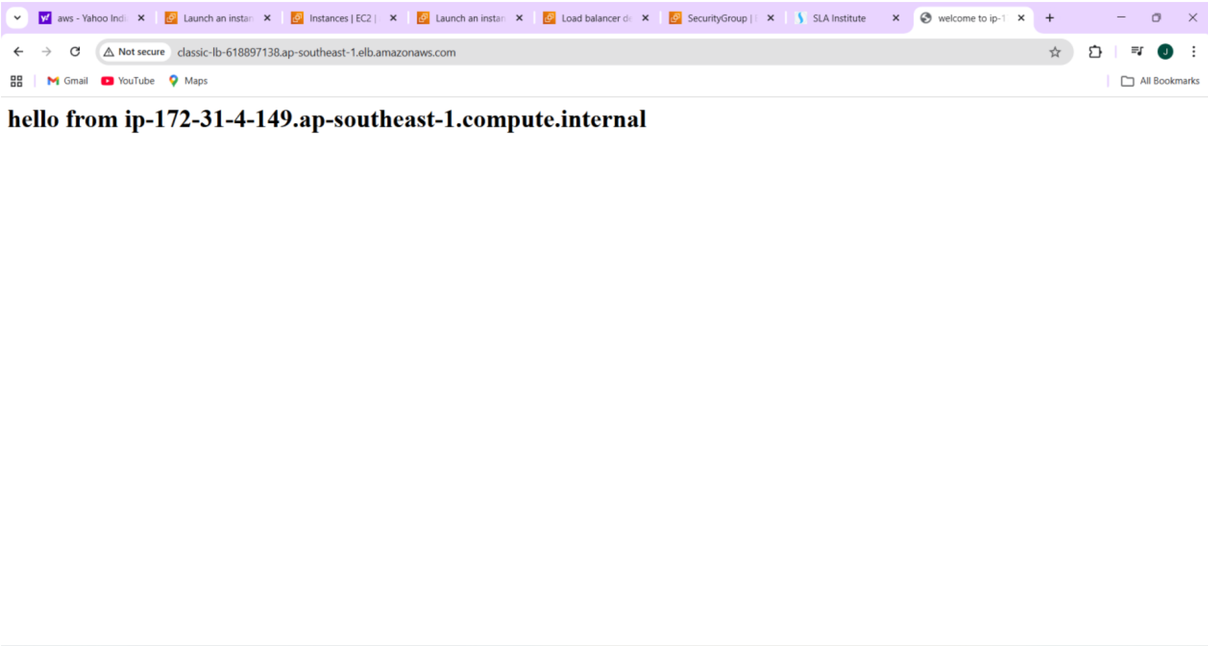
- Go to **Load Balancer details**
- Copy the **DNS name**



Screenshot: Access Application

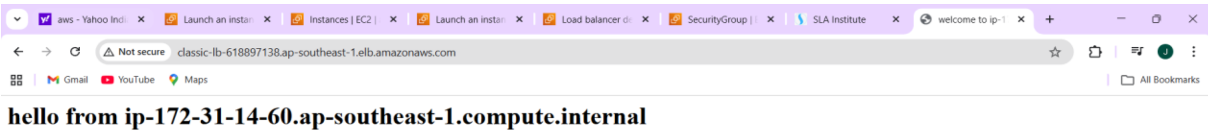
- Paste the DNS name in browser

- Press Enter
- Verify the application is accessible



Screenshot: Verify Load Balancing

- Refresh the browser page multiple times
- Observe that different instance hostnames are displayed
- Verify traffic is distributed between both instances



APPLICATION LOAD BALANCER (ALB)

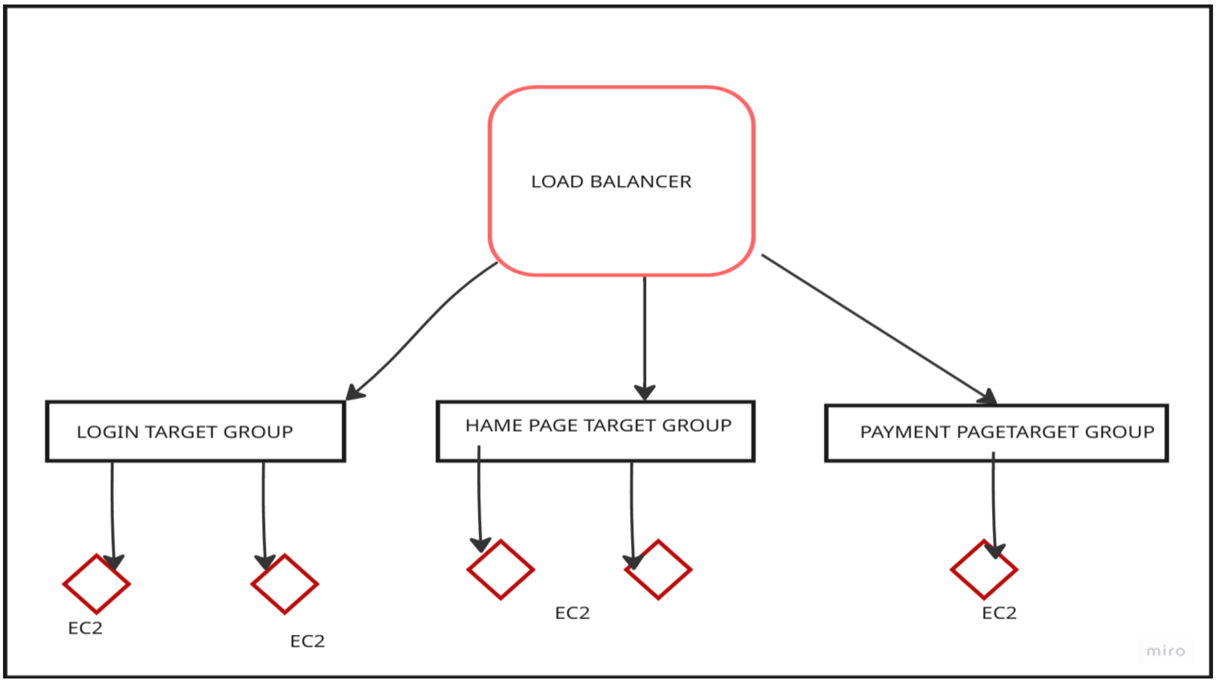
An Application Load Balancer in **Amazon Web Services** is a **Layer 7 load balancer** that distributes incoming HTTP and HTTPS traffic across multiple targets based on request content such as URL paths and hostnames.

Target Group

A Target Group is a logical group of resources (such as EC2 instances) that receives traffic from a load balancer and distributes it based on defined rules and health checks.

EC2 Instance

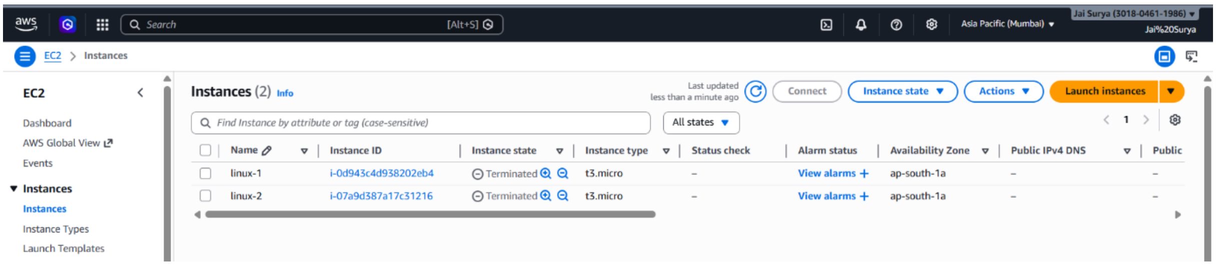
An EC2 instance is a virtual server in AWS used to run applications, which can be registered as a target to receive traffic from a load balancer.



HANDS-ON STEPS FOR APPLICATION LOAD BALANCER (ALB)

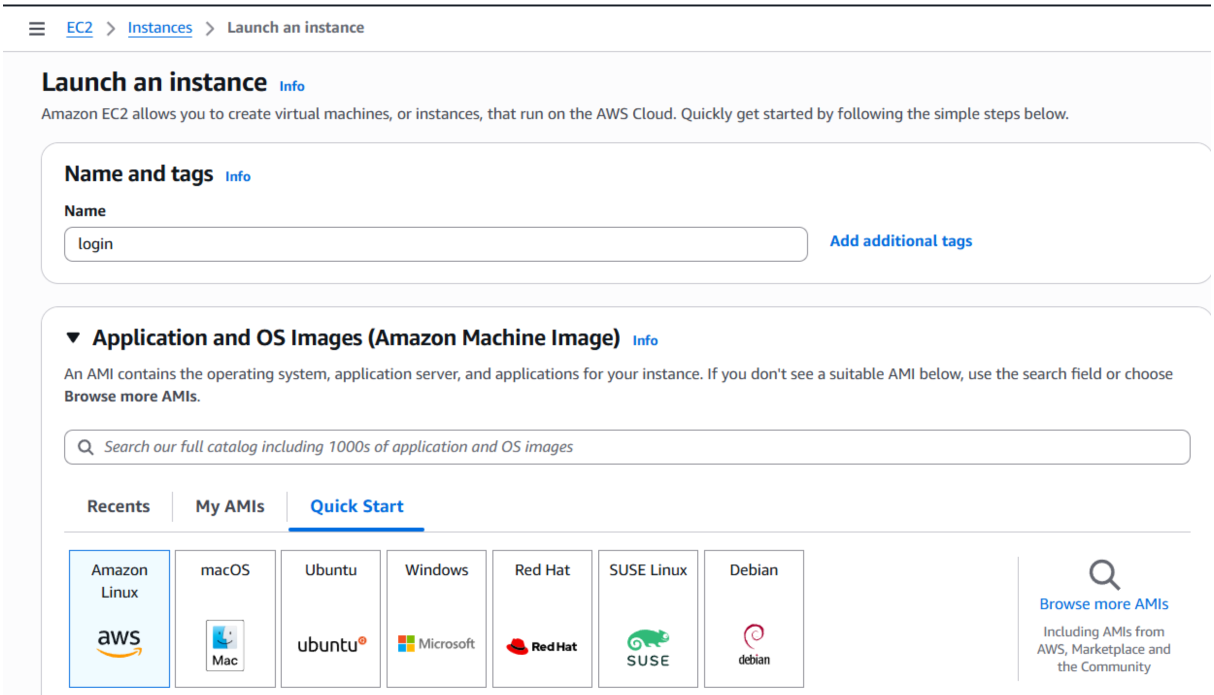
Screenshot: Launch Instance

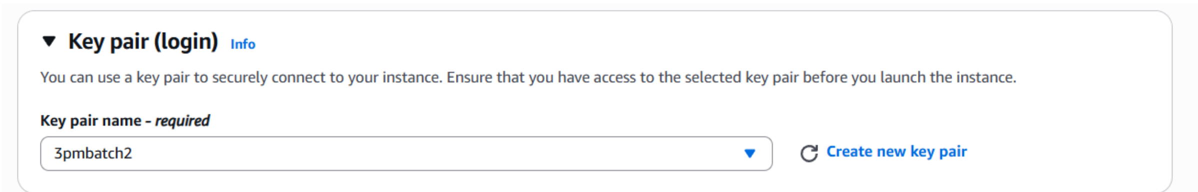
- Go to Amazon Web Services Console
- Open EC2 Dashboard
- Click Launch Instance



Screenshot: Create Instance for Login Page

- Enter instance name
- Select Amazon Linux AMI



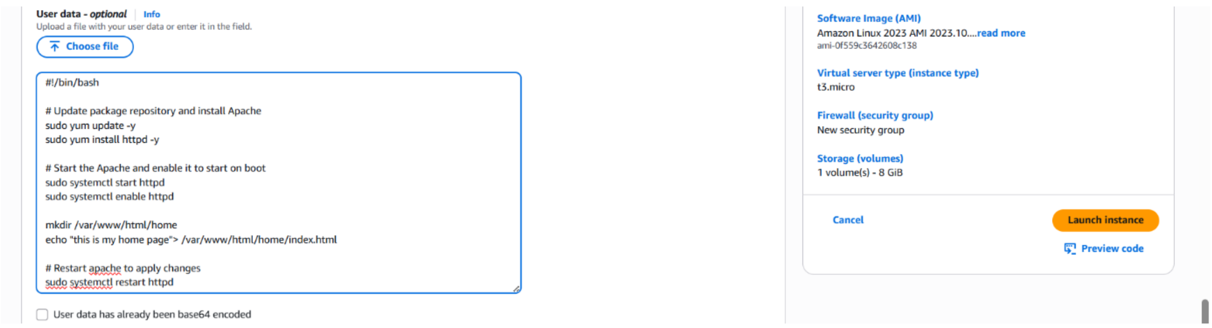


Screenshot: Add User Data (Login Page)

- Go to Advanced Details
- Paste login page script in User Data

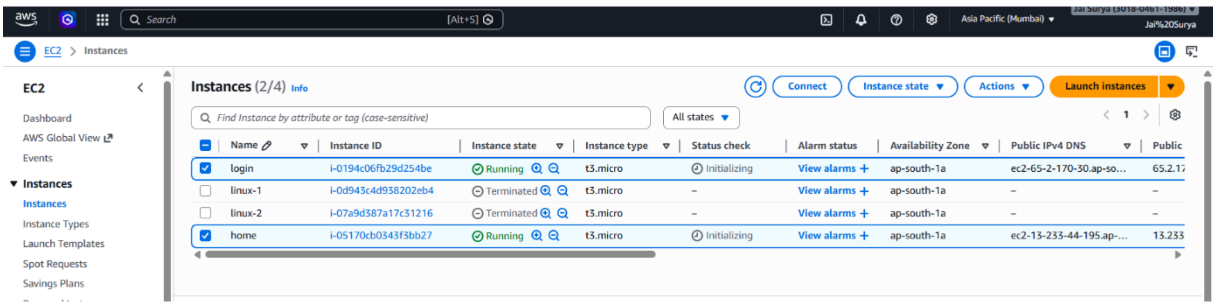
Screenshot: Create Instance for Home Page

- Enter instance name
- Select Amazon Linux AMI
- Paste home page script in User Data



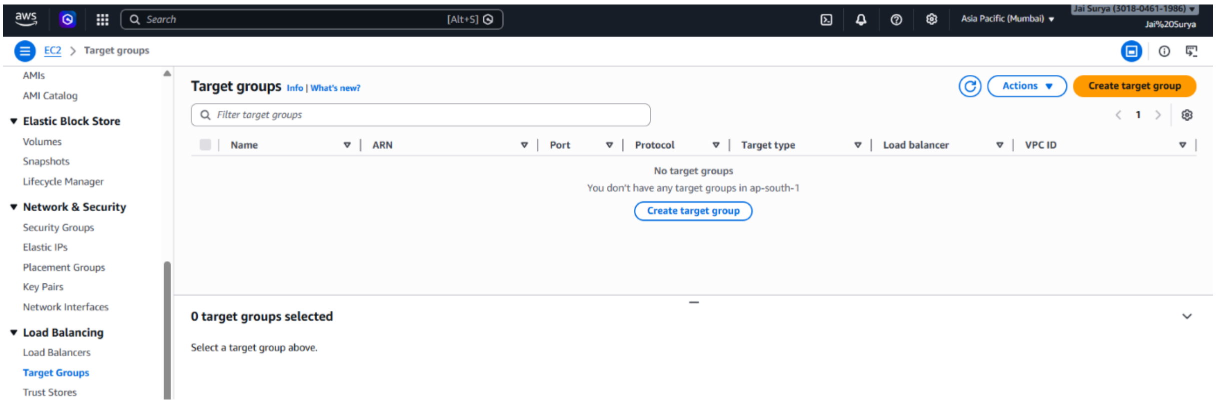
Screenshot: Instances Created

- Verify two instances are created



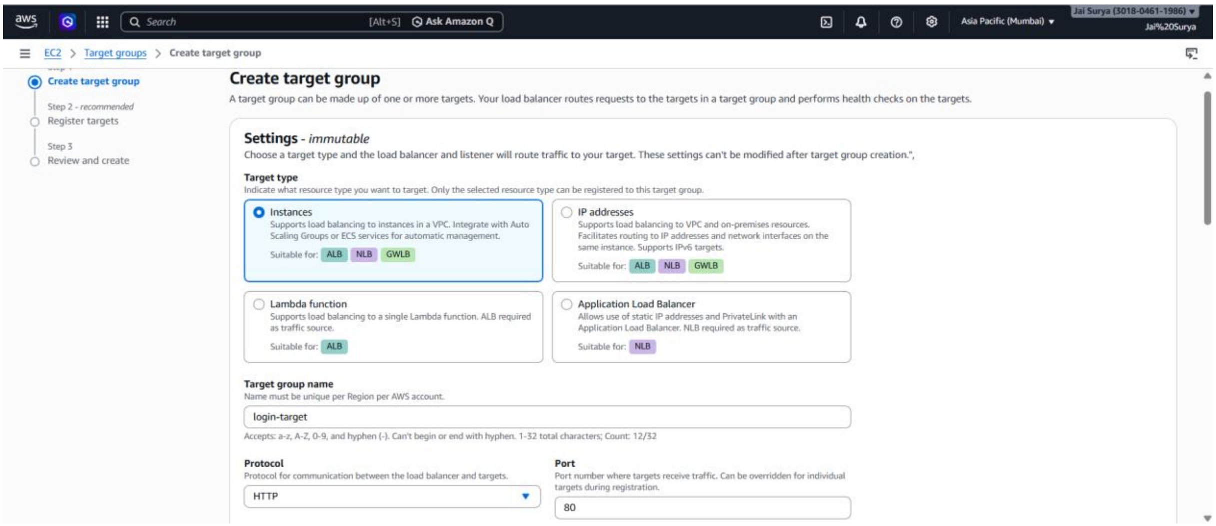
Screenshot: Create Target Group

- Go to **Target Groups**
- Click **Create target group**



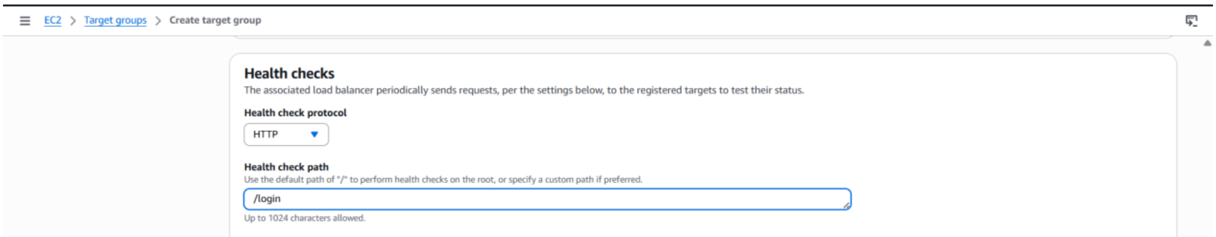
Screenshot: Configure Target Group

- Select **Instances**
- Enter name
- Select protocol **HTTP**



Screenshot: Configure Health Check

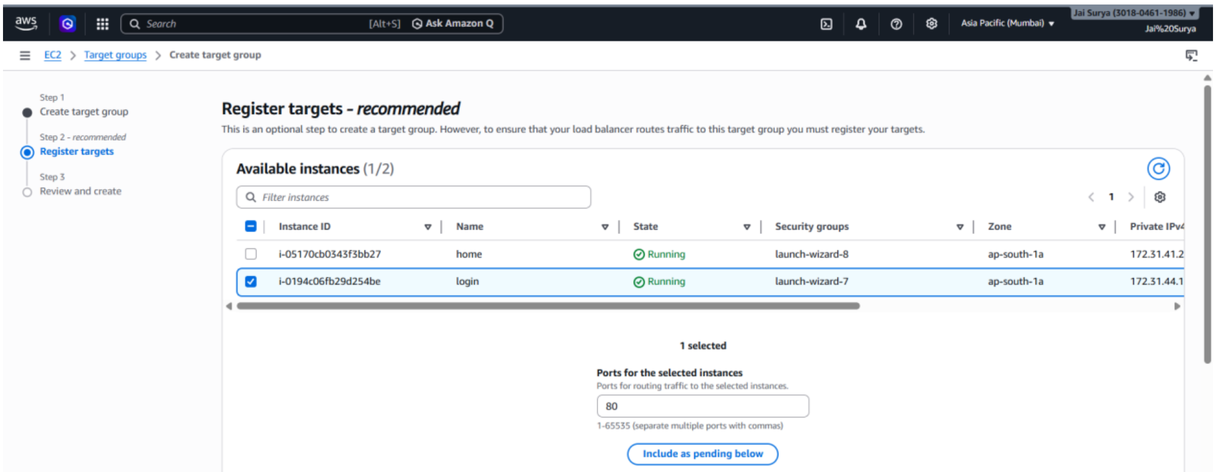
- Set health check path to **/login**

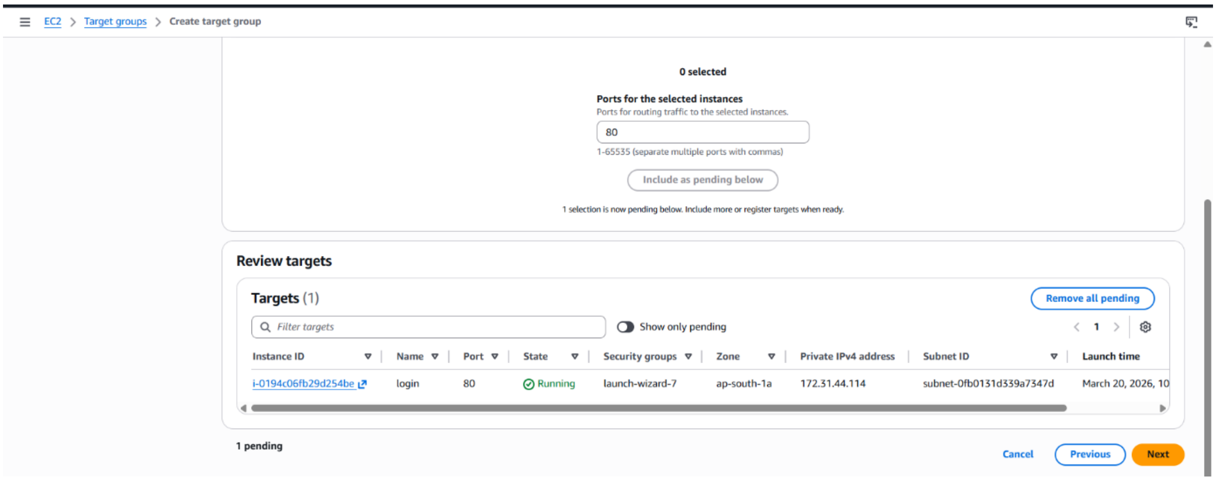


Screenshot: Register Login Instance

- Select login page instance

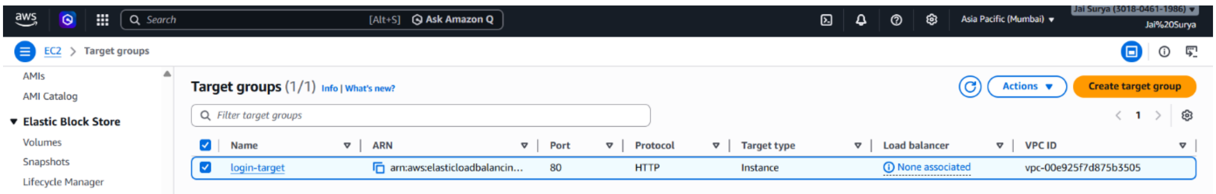
- Click **Add to registered**





Screenshot: Create Target Group

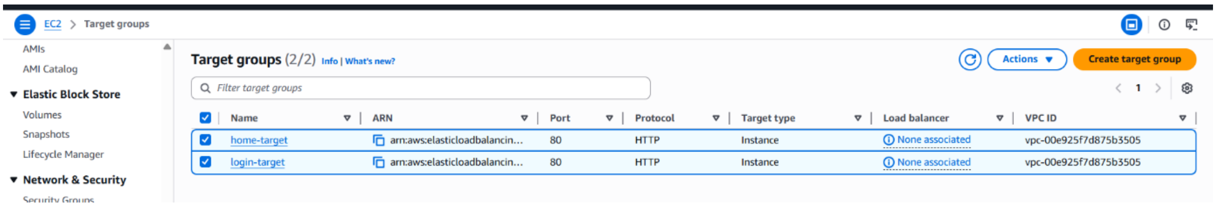
- Click **Create target group**



Screenshot: Create Home Target Group , Target Groups Created

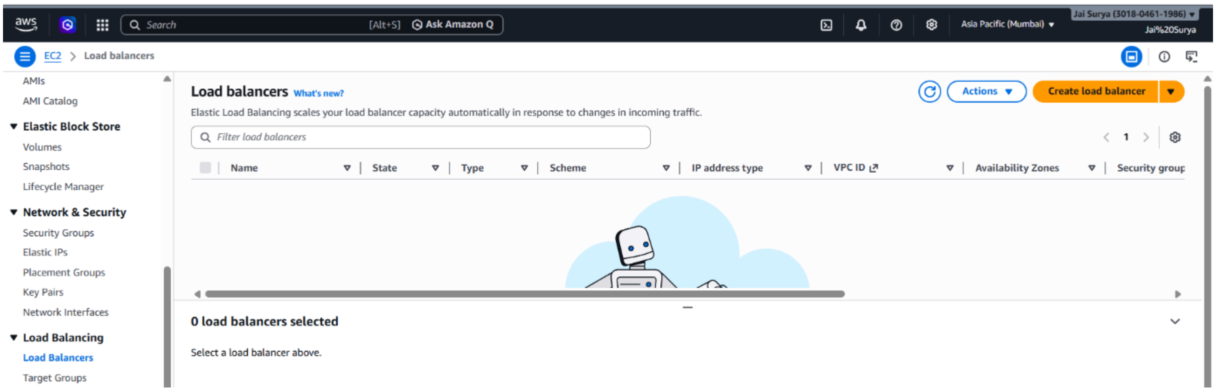
- Repeat the same steps

- Enter name for home target group
- Select protocol **HTTP**
- Set health check path (e.g., **/ home**
- Select home page instance
- Click **Create target group**
- Verify both target groups are created



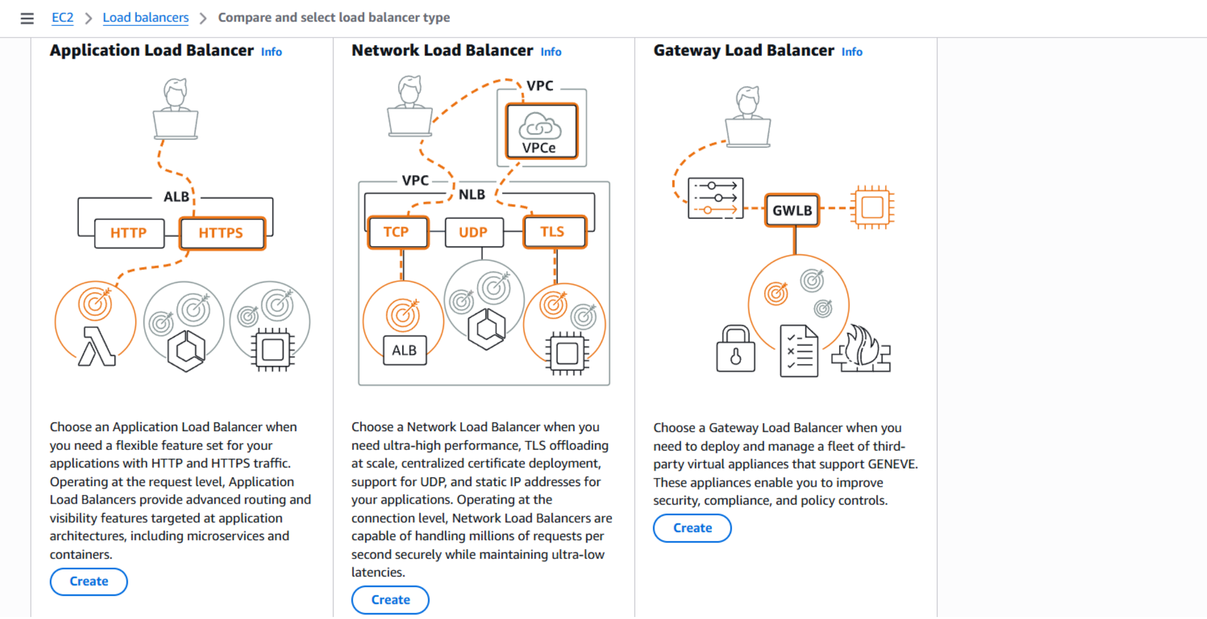
Screenshot: Go to Load Balancer

- Go to **Load Balancers**
- Click **Create Load Balancer**



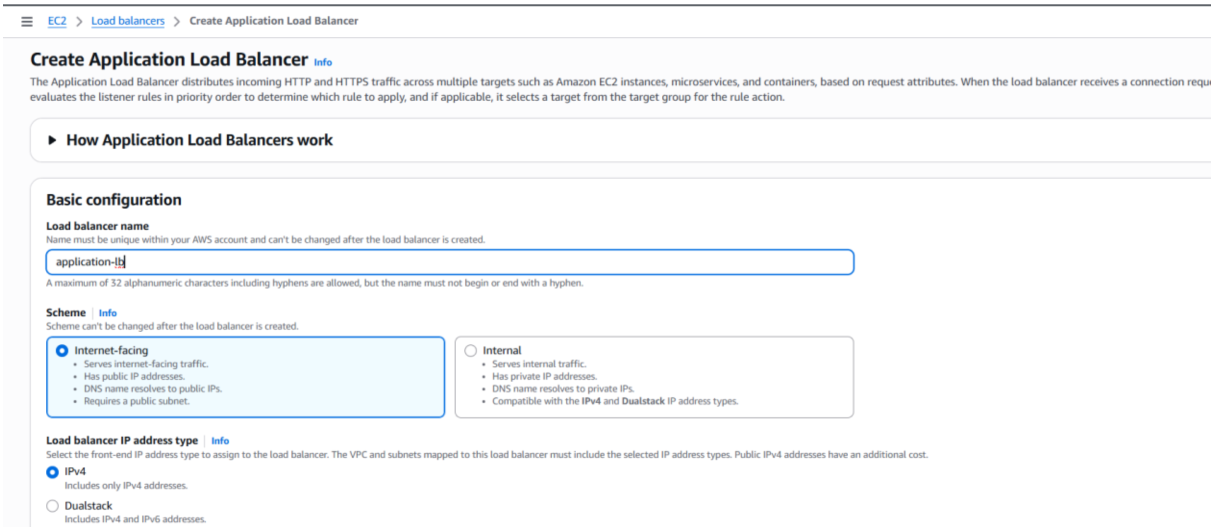
Screenshot: Select Application Load Balancer

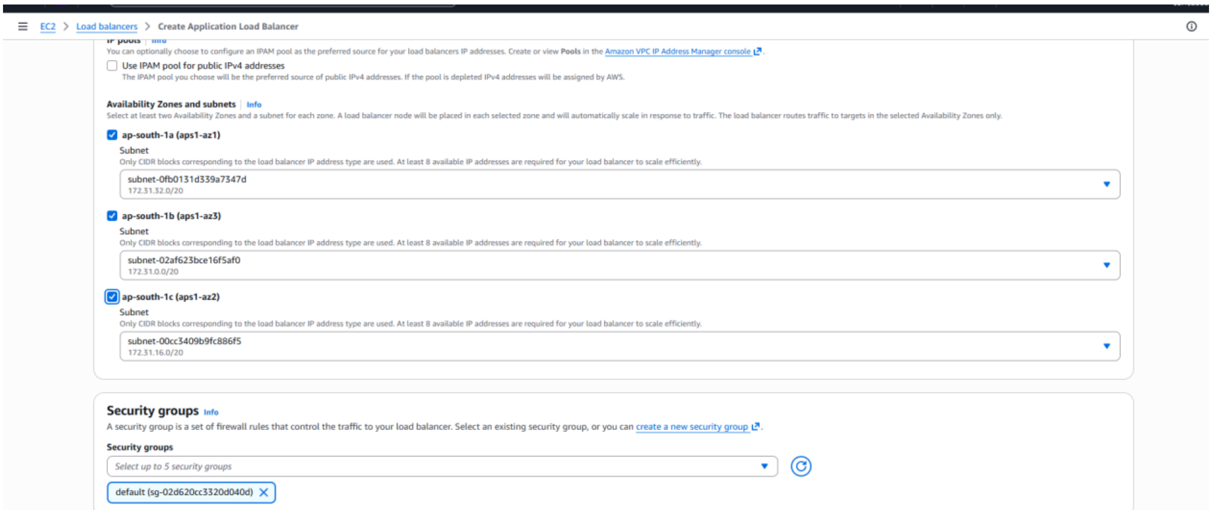
- Choose **Application Load Balancer**
- Click **Create**



Screenshot: Configure Load Balancer

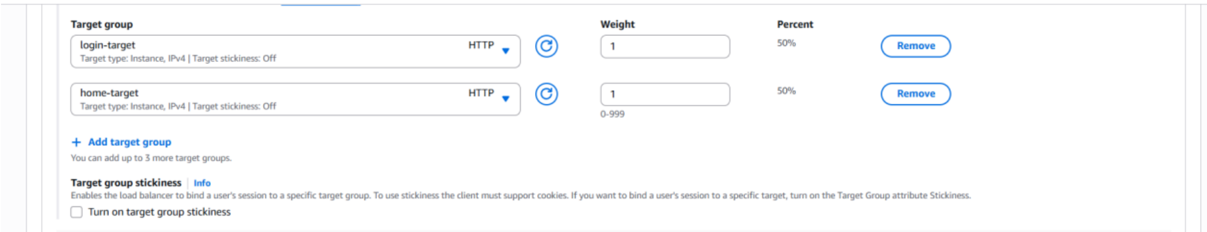
- Enter name
- Select Internet-facing
- Select all subnets





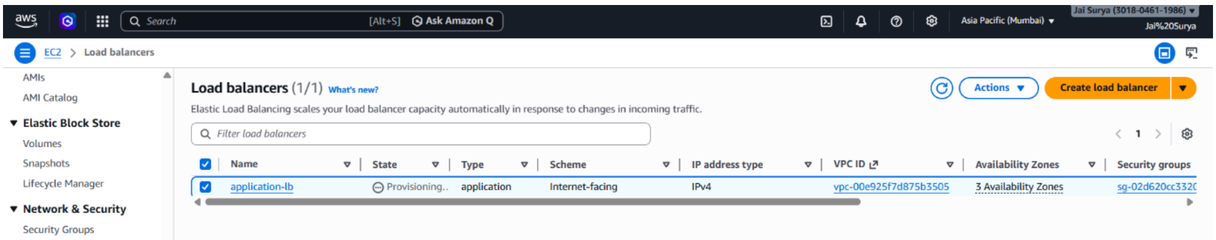
Screenshot: Add Target Groups

- In Target Groups section, click **Add target group**
- Add both **login** and **home** target groups



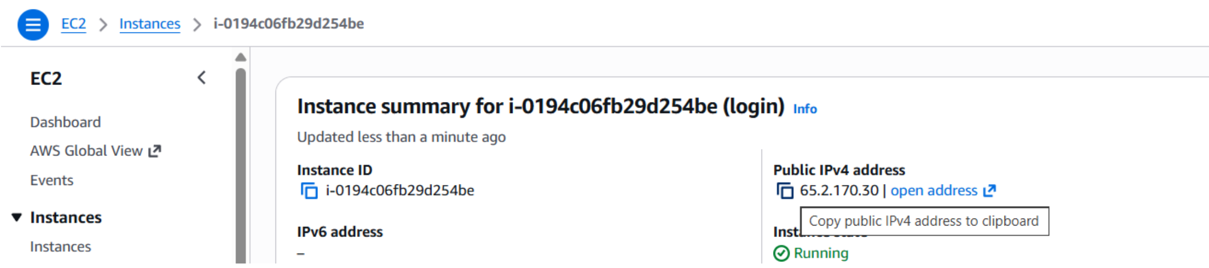
Screenshot: Load Balancer Created

- Verify that the load balancer is created successfully



Screenshot: Access Login Page

- Copy public IP of login instance
- Paste in browser
- Add **/login** to URL
- Press Enter and verify page

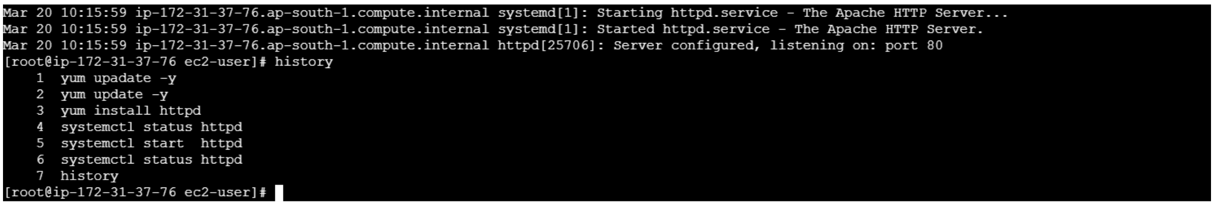


Screenshot: Access Home Page

- Copy public IP of home instance
- Paste in browser
- Add **/home** to URL
- Press Enter and verify page

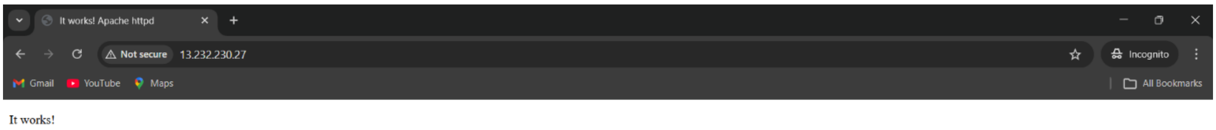
CHANGING PORT NUMBER

Open linux terminal execute the webhosting commands

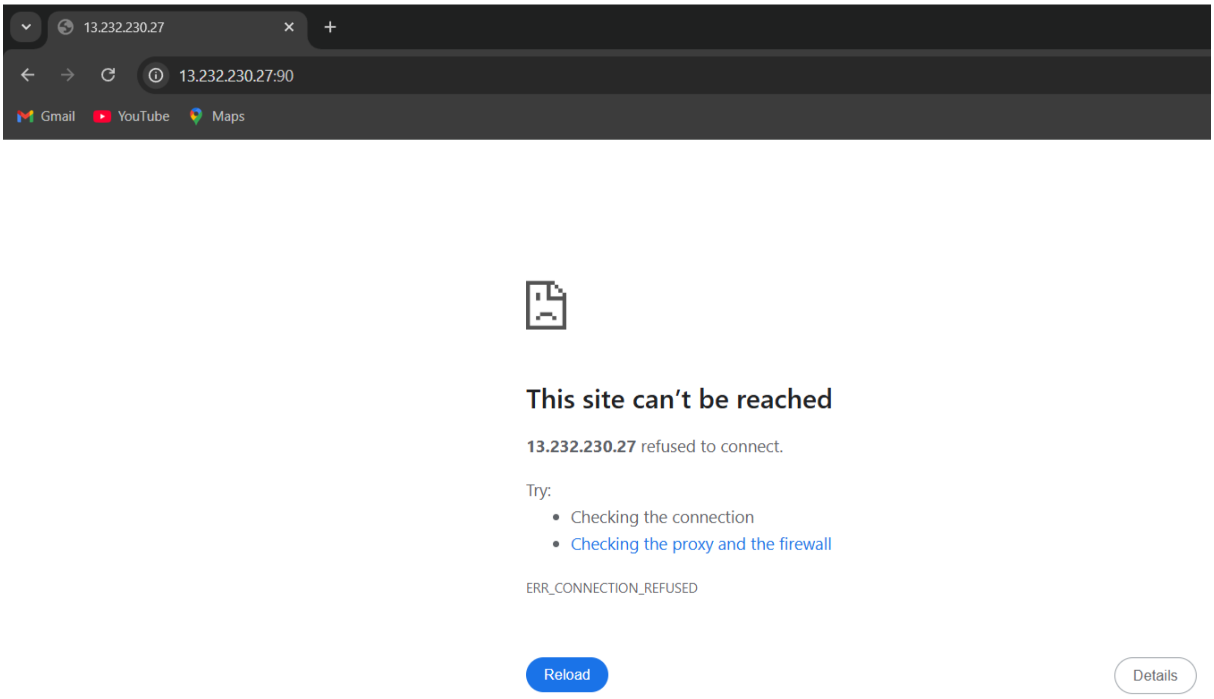


```
Mar 20 10:15:59 ip-172-31-37-76.ap-south-1.compute.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Mar 20 10:15:59 ip-172-31-37-76.ap-south-1.compute.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Mar 20 10:15:59 ip-172-31-37-76.ap-south-1.compute.internal httpd[25706]: Server configured, listening on: port 80
[root@ip-172-31-37-76 ec2-user]# history
    1  yum upadate -y
    2  yum update -y
    3  yum install httpd
    4  systemctl status httpd
    5  systemctl start  httpd
    6  systemctl status httpd
    7  history
[root@ip-172-31-37-76 ec2-user]#
```

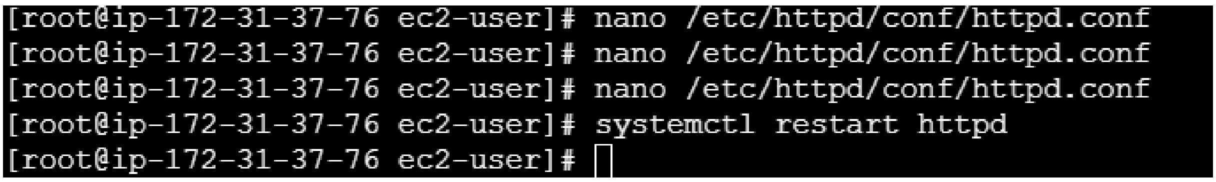
Running in port number 80



Change port number 90 shows error



Execute this command



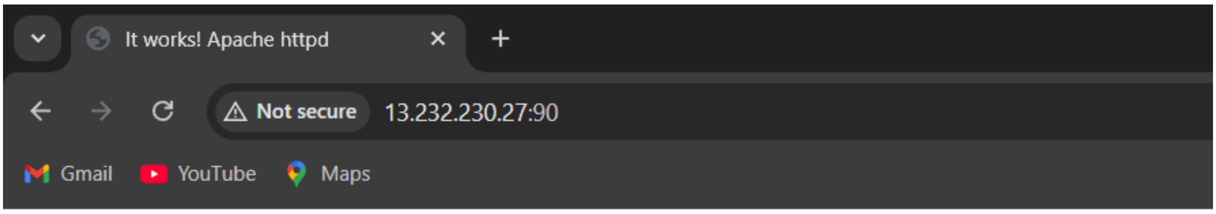
```
[root@ip-172-31-37-76 ec2-user]# nano /etc/httpd/conf/httpd.conf
[root@ip-172-31-37-76 ec2-user]# nano /etc/httpd/conf/httpd.conf
[root@ip-172-31-37-76 ec2-user]# nano /etc/httpd/conf/httpd.conf
[root@ip-172-31-37-76 ec2-user]# systemctl restart httpd
[root@ip-172-31-37-76 ec2-user]#
```

Change port number 90 instead of 80 in listen

```
# available when the service starts.  See the httpd.service(8) man
# page for more information.
#
#Listen 12.34.56.78:80
Listen 90

#
# Dynamic Shared Object (DSO) Support
#
```

Run successfully after changed the port number to 90



It works!